Lab Report Writing Guidelines
(Adapted from Drs. Miller, Moelter and Ríos)

Title:

Name: Partner:

Date Performed:

GOAL

In this section you explain what the experiment is for? It should be short and concise, like an abstract.

- Keep in mind that you can test a theory or a prediction, but you can't prove or verify a theory.
- Copying the Objectives from the write-up/manual will earn no credit.

(There doesn't need to be a title page or blank space here. I only separated the different sections of the report for clarity in this document)

Summary of sections: Title, Goal, Introduction and Background, Procedures and Data, Results and Analysis, Discussion and Conclusions

INTRODUCTION and BACKGROUND

In this section, include the physics that a reader needs to understand this experiment and why we care about this physics. This is looking for the "big picture," not a line-by-line derivation of expressions needed for the data analysis. "The speed of light is a fundamental constant of nature and appears in many branches of physics[1]." Note the superscript referring to one of the items in the Reference list.

- **Equations**: Along with the physics ideas, the background section will most likely contain any equations that will be needed later.
  - Example $$\chi^2 = \sum_{k=1}^{N} \frac{(O_k - E_k)^2}{E_k} \quad (1)$$
  - Upon displaying an equation, indicate what each symbol represents. Note that the equations is numbered for later reference
- **Figures**: This section might include pictures, diagrams, etc. that help explain any theory which is discussed in the report.
- Report writing may not be as much fun as doing the experiment, but it is an essential part of experimental physics. Keep in mind that you're always writing for an audience. In this case, you should assume that your audience is another physicist at your same level. Would another physicist reading your report be able to understand what you did, why you did it, and what you learned?
- The Background is not the same as the Procedure. It is what the experiment is about, not how it is done. While the lab manual has some background information, this section will also depend on reviewing and reading the references.

- This must be in your own words. Not only will no credit be awarded for copying sentences from the lab manual, doing so will result in a substantial penalty.

Conceptual **Q**uestion 1 (CQ1):	**What are we measuring?**
Use these conceptual questions to prompt your thoughts and discussion in the **Introduction and background** section of your lab report.	What concepts and principles are necessary to generate explanatory or predictive models. You should think broadly about laws, equations, and theories. In narrative form, present a thread of physics logic that describes why the experiment might succeed in what it is supposed to measure. These are *the concepts*. In the thread, sprinkle in references to established laws of physics, which will likely use equations and present important quantities (or variables). Your goal is to guide the reader as to the most relevant physics that needs to be established to understand the measurements and data you will present in your technical report.

PROCEDURES

In this section, describe your experimental procedures in paragraph form. It's useful to be very specific, such as "In order to determine whether X is true, I used a supersonic widget. I measured a, b, and c. From these, using the quantum widget theory, I'll be able to calculate Y. I'll know whether X is true or not based on whether Y is positive or negative."

The procedure *should **not** be a set of instructions or a list of what you did*. The key to writing a quality procedure section is that another scientist reading your report should be able to determine (roughly) how you went about doing the experiment (without being overly "literal"). A block diagram showing how different parts of the experiment are related to each other is essential to include. There should always be one to summarize the whole experiment. Other figures are sometimes helpful, including schematics for electrical circuits.

- **Diagrams/Figures:** A block diagram can be helpful. Figures should be numbered and need to include an informative caption.

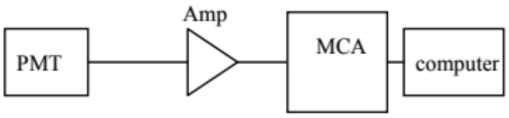

Figure 1. Experimental design. Particles are detected by the PMT, the signal is amplified and the spectrum determined by the MCA.

Conceptual **Q**uestion 2 (CQ2):	**How are we measuring it?**
Use these conceptual questions to prompt your thoughts and discussion in the **Procedure and Data** section of your lab report.	Discuss which experimental parameters are connected to the laws, equations, and theories you discussed in the last section. Be sure to discuss which parameters will be measured by you in lab; which will be calculated; and which will be extracted from data fitting. Summarize what needs to be measured, calculated, or extracted from a fit for your Intro/background to work. Discuss exactly what quantities the equipment allows you to measure, and how it does so. Discuss how the measurable quantities link to the quantities you wrote about in CQ1. If you find yourself introducing equations here that are not in CQ1, that should be a hint to you that your CQ1 answer is incomplete.

RESULTS and ANALYSIS

In this section, you should include any data along with full-featured plots that visualize and analyze your measured data and address any prompt questions asked in the lab manual.

- **How you did the analysis**: What data did you use and what did you do with it. Your analysis needs to discuss how and why you are using the measured values. References to your Introduction or textbook equations is appropriate, to explain what you're doing, but you do not have to justify them. What you want to avoid is "black magic" analysis with equations coming out of nowhere, without explanation, and a bunch of numbers are pushed around with no discussion. "We took the distance and time data plotted in Figure 3 along with the best-fit line. The fitting parameters were used to determine the speed of light." (We don't need to know which language/environment used.)
- **Tables:** In many cases your measured data can be helpfully displayed in tables, such as the example shown below as Table I. You should include a table number and a caption.

Table I. Radius of curvature of electron path for different accelerating voltages.

Voltage	1000 V	1500 V	2000 V
Sample	r (mm) ± 0.3 mm	r (mm) ± 0.4 mm	r (mm) ± 0.5 mm
A	34.5	45.2	56.3
B	37.6	48.9	55.9

Be sure to always include information about the uncertainties in your data. You can do this either by writing 20.2 ± 0.3 cm or, if the data is in columns, writing (in an obvious location) "all temperatures ±1°C."

- **Full-featured plots include**
  - Title
  - Figure caption
  - Clear, labeled axes with units ("*I* (mA)")
  - Data points with error bars (X and/or Y)
  - Trendline (as appropriate)

    - Equation of trendline parameters with uncertainties
    - Legend (if appropriate)
- Compare to accepted value
    - Clearly state your measured value in relation to the accepted value.
    - Your use of statistics and analysis of uncertainties goes here. You will often use two different analysis methods (linear fit and SDOM) and compare them. It's important to learn to be "reasonable" with uncertainty analysis without beating the issue to death. Focus on the major sources of error or uncertainty. Don't do some horrendous calculation with dozens of partial derivatives. You should be able to recognize the one or two major sources of uncertainty and focus on those. Show and explain your uncertainty analysis.

Your report should also discuss the significance of what you've learned while stating any assumptions you used.

Conceptual **Q**uestion 3 (CQ3):	**How is the measurement or the model limited?**
Use these conceptual questions to prompt your thoughts and discussion in the **Results and Analysis** section of your lab report.	Discuss what simplifications, limitations, and *assumptions* you have made in either the concepts and principles, or in determining the experimental parameters. You may ask yourself something like, "In what regimes of my relevant experimental parameter does this equation hold? Why does it hold?"

DISCUSSION/CONCLUSION

Have the goals of the experiment been satisfied? If not, why not? Getting the "right answer" is far less important than showing that you understand what happened in the experiment. Be sure to restate your main result. Some of our experiments cannot give particularly high accuracy, but a reasonable discussion of the limitations will show your understanding of the critical features of the experiment. A short discussion of any interesting implications of the experiment is in order here.

Do not state that "our value of x was pretty close to the accepted value of x". Be quantitative. "The accepted value of x lies within 2σ of our mean measured value of x" or "Our measured value is 6% different than the accepted value" or something similar.

Conceptual **P**rompt 4 (CQ4):	**Summary**
Use these prompts to prompt your thoughts and	Finally, synthesizing everything you've thought and written about above, write 2-3 sentences summarizing the connection

discussion in the **Discussion/Conclusion** section of your lab report.	between what you are measuring, how you are measuring it, and why you are measuring it. Your answer should include elements from the three previous questions (CQ1-3).

REFERENCES

References should be to materials other than the lab manual. Do not reference the lab manual. Instead, finding and listing an appropriate textbook, another book, or paper would be appropriate. It should be a reference to something that is published.

References Example:

1. A.C. Melissinos and J. Napolitano, Experiments in Modern Physics, 2nd ed. (Academic Press, NY, 2003).
2. Thomas J. Bensky and Matthew J. Moelter, "Computational problems in introductory physics: Lessons from a bead on a wire," Am. J. Phys. **81**, 165 (2013).
3. C.E. Miller, J. Majewski, T. Gog, and T.L. Kuhl, "Characterization of Biological Thin Films at the Solid-Liquid Interface by X-Ray Reflectivity" Phys, Rev. Lett. **94**, 238104 (2005).

Dr. Miller's disclaimer (opinion): High quality and effective scientific writing (argumentation/communication) is challenging to produce. It is an art form which takes practice. It is a powerful skill that is entirely independent from performing quality measurements. In this course, you will be performing measurements that have been repeated many times before by countless, unnamed experimenters. These countless measurements are tests to a theory/model that have accepted values to which you will compare your results in your reports. Now imagine trying to test new theories of physics. The communication of those ideas to a scientific audience is paramount… so the next scientist can verify your findings.

Example (mostly from Dr. Rios)

Here is a short example of how one might answer the above Conceptual Questions for a hypothetical experiment using a pendulum to determine *g*. This is a rephrasing of what I would think to myself to start writing. The answers are very brief here, because the experiment is straightforward. You should elaborate a bit more for your QLab reports

Q 1. What are the main principles and concepts I need to describe to fully contextualize how and why I'm measuring g with a pendulum?

The foremost principle we draw from in this lab is that of a harmonic oscillator--the motion of a pendulum is oscillatory with a characteristic frequency. By recognizing that the pendulum swing can be modeled using Hooke's Law, $F = -kx$ (where k is the spring constant and x is the position of the mass), we can equate Hooke's Law to Newton's Second Law so that $ma = -kx$. We can solve this differential equation and find that the period of a pendulum can be written as

$$T = 2\pi\sqrt{\frac{L}{g}}$$

where L is the length of the pendulum arm and g is the acceleration due to gravity. This equation is an approximation; it is the first term of a Taylor expansion of the solution to the differential equation of a mass on a spring.

Q 2. ***Okay, now that I have described the physics ideas we need, what did we measure that is related to those ideas?***

The equation above predicts the period of a pendulum of a given arm length. Thus, we can plot a series of *squared* Ts against L to obtain

$$T^2 = \frac{4\pi^2}{g}L$$

and fit to a simple linear equation, $y = mx$. When T^2 is plotted against the length, L, (with a constant), the extracted parameter of the slope yields g. The two measurable experimental parameters in this lab are the period, T, and the length of the pendulum, L.

Q3. Okay, this one is tricky! I simplified the model quite a bit, which means I am also implicitly assuming the experiment itself can also be simpler. I must also be able to justify why any assumption I make is valid, or how the data will let me know that it is valid. Let me try to be extra clear.

As mentioned, the equation

$$T = 2\pi\sqrt{\frac{L}{g}}$$

is the first term of a Taylor expansion. We can use this equation if **we assume the angle of displacement is small**. We can define "small" by considering the second term in the equation:

$$T = 2\pi\sqrt{\frac{L}{g}}\left(1 + \left(\frac{1}{16}\theta^2\right)\right)$$

For this experiment, we will assume that θ (initial displacement angle) is small if it is less than 10 degrees. I will know if this assumption is valid if I obtained linear data. If I start to get nonlinearities, that will indicate to me that perhaps the angles are not small enough. Another implicit assumption here is that the exact angle does not matter if it is small enough. That can be justified by inspecting, again, the model equation---it does not contain an angle term. **A second assumption inherent in the equation is that the string holding the weight is "massless."** In this experiment, we define "massless" as the string's mass is much, much smaller than the mass at the end of the pendulum arm, not a literal zero mass. This is an important assumption because we want to only consider forces involved in Hooke's Law and Newton's Second law, as discussed in the first conceptual question; an additional force from the tension of the string would make solving our differential equation more complicated. We can ensure this assumption will hold by using a heavy, bronze mass at the end of one light string. Lastly, additional considerations like resistance from the air as the pendulum moves could affect the measurement of *T*, the period. However, we think those effects will be small if the first two limitations are addressed and will probably have small contributions to the random errors in the fit. Inspection of the uncertainty at 68% confidence will let us know if we need to account for these effects.

Q4. Almost done! Now I must create some cohesiveness around all the ideas I presented above.

We are measuring the acceleration due to gravity, *g*, using a simple equation that is derived from considering the simple harmonic motion of a pendulum. Deviations from an accepted value could be due to a large displacement angle (which is easily remedied by redoing the measurements with smaller angles until the equation is valid). Probably the largest source of error will be introduced in measuring the length of the pendulum arm, *L*. If we see a large intercept in the linear equation (not expected from the equation), it may be because the effective pendulum arm is shorter or longer than measured.